

Infrared Interferometric Estimation of Silicon Carbide Epitaxial-Layer Thickness

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Abstract

We study the determination of silicon carbide (SiC) epitaxial-layer thickness from infrared reflectance fringes. The original analysis is organized in two stages. A two-beam model first relates the measured reflectance to optical-path difference, refractive index, incidence angle, and layer thickness. The refractive index is treated as dispersive, and the high-wavenumber interval 2500–4000 cm⁻¹ is used because the lattice-absorption region near 800–1000 cm⁻¹ distorts the interference pattern. A second model sums repeated internal reflections to examine when multiple-beam interference materially changes the result. Applied to the measured SiC spectra, the two-beam calculation gives thicknesses of 8.50 and 8.60 μm for incidence angles of 10° and 15°, with reported fit correlations of 99.98% and 99.96%. The multiple-beam calculation gives 8.55 and 8.63 μm for the same two datasets and 3.79 and 3.53 μm for two silicon-wafer datasets. The close agreement between the two optical treatments is an internal consistency check; it is not an independent calibration of absolute thickness accuracy.

Keywords: silicon carbide; epitaxial layer; FTIR reflectance; thin-film interference; multiple reflections

1 Introduction

SiC epitaxy is used in power-electronic and optoelectronic structures, and layer thickness is a basic process parameter. Infrared reflectance provides a non-destructive route to thickness estimation because the air/epitaxial-layer and epitaxial-layer/substrate interfaces generate a phase difference between reflected beams. The resulting fringes encode optical thickness rather than geometric thickness alone, so a usable estimator has to account for refractive index, incidence geometry, and the possibility of repeated internal reflection.

The measurements comprise reflectance spectra acquired at two incidence angles together with two silicon-wafer datasets. The analysis begins with a single-round-trip approximation, derives the thickness estimator for the two SiC measurements, and then extends the optical model to repeated internal reflections as a consistency check. This progression is also consistent with recent FTIR work on SiC, which emphasizes that fringe-spacing methods can be affected by dispersion, absorption, and interface-dependent phase behavior.[1]

2 Two-Beam Optical Model

2.1 Reflection, transmission, and path difference

Let I_0 be incident intensity, R_1 the intensity reflectance at the air/epitaxial-layer interface, and R_2 the corresponding reflectance at the epitaxial-layer/substrate interface. In the first approximation only the direct surface reflection and the component that traverses the layer once in each direction are retained. Thus

$$I_1 = I_0 R_1, \quad I_3 = (1 - R_1) I_0 R_2. \quad (1)$$

The two components are treated as spatially overlapping. With layer thickness d , refractive index n , external incidence angle θ_0 , and refracted angle θ_t , the project writes the optical-path difference as

$$\Delta L = \frac{2dn}{\cos \theta_t} + \frac{\lambda}{2} - 2dn \tan \theta_t \cos \theta_0. \quad (2)$$

The $\lambda/2$ term accounts for the reflection phase reversal in the adopted convention. The combined intensity is

$$I = I_1 + I_3 + 2\sqrt{I_1 I_3} \cos\left(\frac{2\pi\Delta L}{\lambda}\right). \quad (3)$$

After division by I_0 , the measured reflectance is therefore a slowly varying envelope multiplied by an approximately sinusoidal phase term.

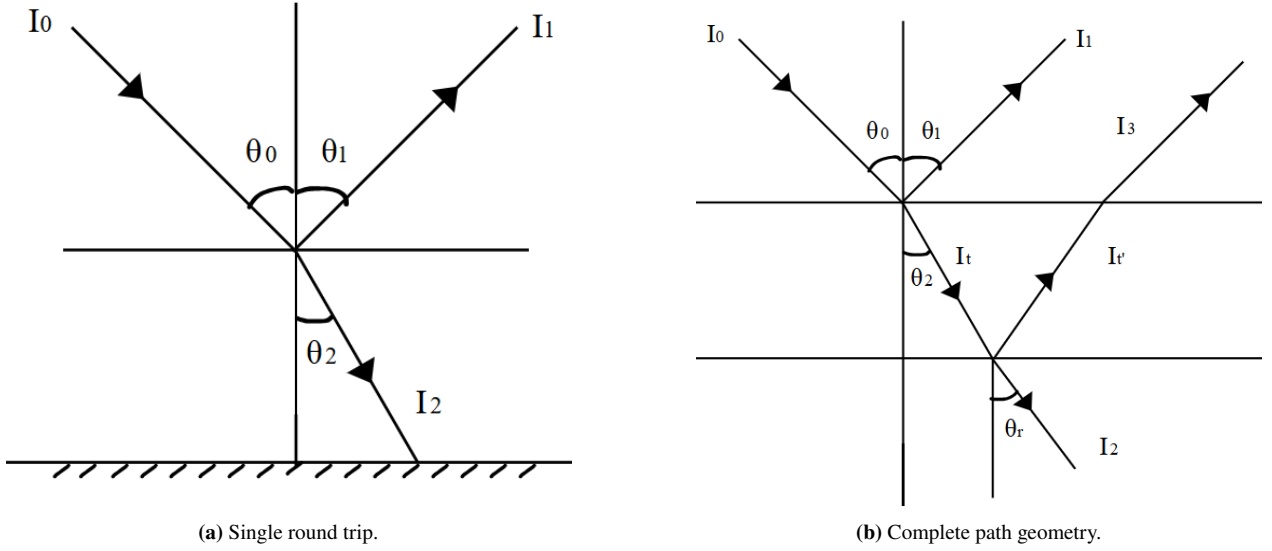


Figure 1: Optical-path diagrams from the original project. These schematics contain no language-dependent labels and are retained directly.

2.2 Wavenumber form

Writing wavenumber as $s = 1/\lambda$ and using Snell's law, the angular factor used in the source calculation can be written

$$n \left(\frac{1}{\cos \theta_t} - \tan \theta_t \cos \theta_0 \right) = \frac{2n^2 - \sin 2\theta_0}{2\sqrt{n^2 - \sin^2 \theta_0}}. \quad (4)$$

Hence the oscillatory term has phase proportional to sd . If R_1 , R_2 , and n change slowly over one fringe, adjacent extrema provide a local thickness estimate.

3 Methodology

3.1 Dispersion and Spectral-Window Selection

The project does not assume a constant refractive index over the entire infrared spectrum. It uses the Sellmeier-type relation

$$n^2 = 1 + \frac{0.20075\lambda^2}{\lambda^2 + 12.07224} + \frac{5.54861\lambda^2}{\lambda^2 - 0.02641} + \frac{35.65066\lambda^2}{\lambda^2 - 1268.24708}. \quad (5)$$

Within the 2500–4000 cm^{-1} window, neighboring fringe extrema are separated by roughly 120–135 cm^{-1} in the source analysis, and the calculated change in n between neighboring extrema is at most about 1%. The calculation therefore treats n as locally constant over one fringe while retaining its spectral dependence over the wider band.

The low-wavenumber region is deliberately excluded. Around the SiC lattice-vibration absorption band, roughly 800–1000 cm^{-1} in the analysis, both refractive index and extinction coefficient vary rapidly and the regular interference fringes are strongly distorted. Thickness extraction is therefore performed only on the high-wavenumber interval where extrema remain identifiable.

3.2 Extrema selection and polarization check

Peak positions are noisy. The project selects the first and last reliable extrema in the analysis interval and uses approximately equally spaced intermediate locations, reducing sensitivity to a single local peak displacement. If neighboring extrema are (x_1, y_1) and (x_2, y_2) in wavenumber-reflectance coordinates, the two-beam equations used in the source are

$$R_1 + R_2(1 - R_1) = \frac{y_1 + y_2}{2}, \quad (6)$$

$$2\sqrt{R_1 R_2(1 - R_1)} = \frac{|y_1 - y_2|}{2}. \quad (7)$$

For the modest incidence angles in the data, the project also checks the Fresnel polarization correction. At 15° and $n \approx 2.5$, it obtains about 19.41% for s polarization and 17.35% for p polarization, while the unpolarized approximation is about 18.37–18.38%. On that basis the subsequent thickness calculation neglects polarization splitting.

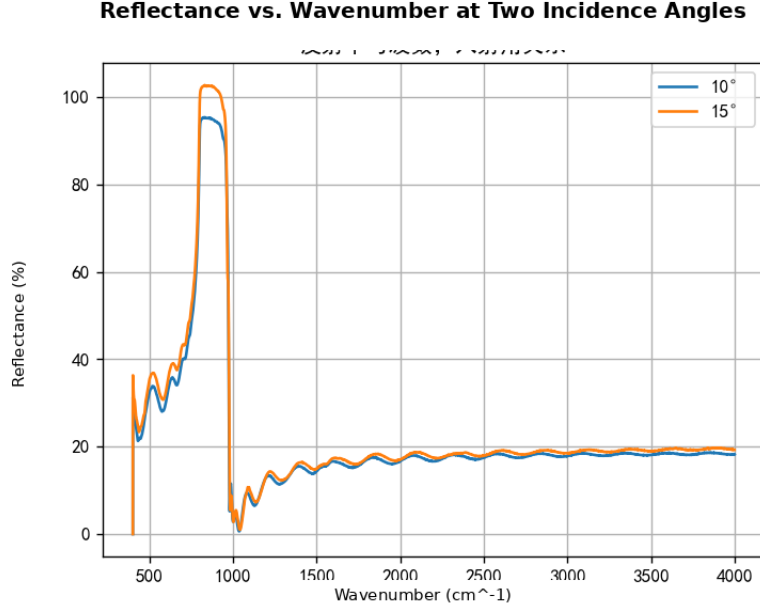


Figure 2: Measured reflectance spectra used in the original analysis.

Using $R \approx [(n - 1)/(n + 1)]^2$ and the extrema equations, the local refractive index is written as

$$n = \frac{2 + \sqrt{y_1} + \sqrt{y_2}}{2 - \sqrt{y_1} - \sqrt{y_2}}, \quad (8)$$

and the corresponding thickness estimator is

$$d = \frac{1}{4(x_2 - x_1)} \left[\frac{2n^2 - \sin^2 2\theta_0}{2\sqrt{n^2 - \sin^2 \theta_0}} \right]^{-1}. \quad (9)$$

This is the estimator applied to the first two datasets.

Multiple-Reflection Model. Repeated internal reflections become relevant when the two interfaces are sufficiently parallel, absorption is small, and the epitaxial layer and substrate have enough refractive-index contrast to produce appreciable reflection at the buried interface. Let r_1, r_2 and t_1, t_2 denote amplitude reflection and transmission coefficients. The transmitted field is a geometric series,

$$\tilde{U}'_k = (t_1 t_2)(r_1 r_2)^{k-1} A_0 e^{i(k-1)\delta}, \quad \delta = \frac{2\pi}{\lambda} 2nd \cos \theta_t, \quad (10)$$

so that

$$\sum_{k=1}^{\infty} \tilde{U}'_k = \frac{(t_1 t_2) A_0}{1 - r_1 r_2 e^{i\delta}}. \quad (11)$$

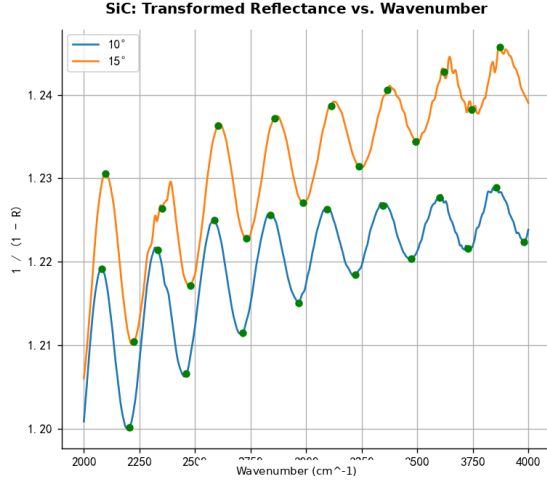
The intensity transmission and reflection become

$$\frac{I_T}{I_0} = \frac{(1 - R_1)(1 - R_2)}{1 + R_1 R_2 - 2\sqrt{R_1 R_2} \cos \delta}, \quad (12)$$

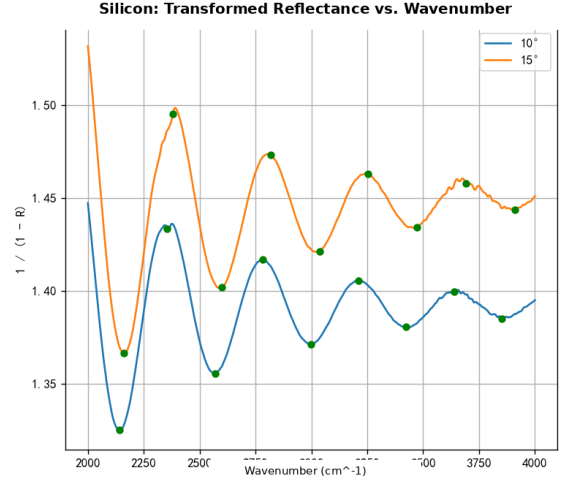
$$\frac{I_R}{I_0} = 1 - \frac{(1 - R_1)(1 - R_2)}{1 + R_1 R_2 - 2\sqrt{R_1 R_2} \cos \delta}. \quad (13)$$

The source project uses a 5% change in recovered thickness as a practical threshold for deciding whether multiple-beam interference is consequential. This is a modeling convention for the project, not a general metrological standard.

The transformed $s-1/(1-R)$ plots in the study are then used to recover $R_1 R_2$, the local refractive index, and finally d . The algebra leads back to a thickness expression with the same phase-spacing structure as the two-beam estimator, but with the interface reflectances inferred from the Airy-type response rather than the simpler envelope equations.



(a) SiC dataset.



(b) Silicon dataset.

Figure 3: Transformed spectra used for the multiple-reflection calculation.

Table 1: Thickness estimates from the two-beam and multiple-reflection calculations.

Dataset	Model / incidence	Thickness (μm)	Reported correlation
SiC 1	two-beam, 10°	8.50	99.98%
SiC 2	two-beam, 15°	8.60	99.96%
SiC 1	multiple-beam	8.55	99.91%
SiC 2	multiple-beam	8.63	99.83%
Silicon 3	multiple-beam	3.79	99.97%
Silicon 4	multiple-beam	3.53	99.96%

4 Results

The two SiC estimates differ by only $0.10 \mu\text{m}$ between the two incidence angles. Introducing the repeated-reflection treatment changes the estimates by 0.05 and $0.03 \mu\text{m}$, respectively. For these data, the multiple-beam correction is therefore small relative to the layer thickness. The additional silicon-wafer calculations give substantially thinner layers, 3.79 and $3.53 \mu\text{m}$.

The project reports correlations close to unity for the transformed relationships. These numbers support internal fit consistency, but they should not be read as 99.9% measurement accuracy. Absolute accuracy would require an independent thickness reference and an explicit uncertainty budget.

Multiple-Reflection Criterion. The original derivation goes beyond writing the Airy denominator: it asks when the difference between a two-beam interpretation and a multiple-beam interpretation is large enough to matter for thickness recovery. This is useful because the mere existence of repeated internal reflections does not imply that they change the estimated thickness appreciably. The analysis uses a 5% change in recovered thickness as its working criterion for a material multiple-reflection effect.

For fixed interface reflectances over a short spectral interval, the extrema of the multiple-reflection response can be written in terms of R_1 and R_2 as

$$y_1 = \left(\frac{\sqrt{R_1} + \sqrt{R_2}}{1 + \sqrt{R_1 R_2}} \right)^2, \quad y_2 = \left(\frac{\sqrt{R_1} - \sqrt{R_2}}{1 - \sqrt{R_1 R_2}} \right)^2. \quad (14)$$

If these extrema were interpreted with the simpler two-beam formula, the corresponding apparent index n' differs from the underlying index n . The source algebra reduces the comparison to a factor involving $\sqrt{R_1 R_2}$ and obtains the project-specific condition

$$\sqrt{R_1 R_2} < \frac{1}{41} \quad (15)$$

for a thickness change below the adopted 5% threshold. The value $1/41$ should not be read as a universal optical boundary; it follows from the chosen 5% tolerance and the approximations made in the derivation.

Table 2: Optical diagnostics reported with the multiple-reflection estimates.

Dataset	d (μm)	mean n	var.(n)	mean reflectance
SiC 1	8.55	2.47	5.79×10^{-4}	18.0%
SiC 2	8.63	2.53	1.18×10^{-4}	18.8%
Si 3	3.79	3.25	5.66×10^{-4}	28.0%
Si 4	3.53	3.49	9.66×10^{-4}	30.7%

The report then gives silicon as a numerical illustration. Taking $n \approx 3.5$ gives $\sqrt{R_1} \approx 5/9$. Under the project’s calculation, the buried-interface reflectance has to exceed roughly 9/205 before the multiple-beam correction is considered. Expressed as a substrate-index condition, the same calculation gives a threshold near $n_c < 2.29$. This exercise explains why interface contrast, parallelism, and absorption are treated together rather than as independent checkboxes: high finesse requires both appreciable reflection and preservation of coherence over repeated round trips.

4.1 Recovering interface reflectances from transformed extrema

The source calculation uses the transformed ordinate $1/(1 - R)$. For two adjacent extrema (x_1, y_1) and (x_2, y_2) on that transformed curve,

$$\frac{1 + R_1 R_2}{(1 - R_1)(1 - R_2)} = \frac{y_1 + y_2}{2}, \quad (16)$$

$$2 \frac{\sqrt{R_1 R_2}}{(1 - R_1)(1 - R_2)} = \frac{|y_1 - y_2|}{2}. \quad (17)$$

Defining

$$A = \left(\frac{y_1 + y_2}{y_1 - y_2} \right)^2, \quad (18)$$

the physically admissible solution in the report is

$$B = R_1 R_2 = 2A - 1 - 2\sqrt{A^2 - A}, \quad (19)$$

where the alternative root is rejected because it would give $R_1 R_2 > 1$. The remaining symmetric combination of R_1 and R_2 is then obtained from the first extrema equation, after which the two interface reflectances are recovered as roots of a quadratic. The larger root is assigned to the air/film interface following the ordering assumption used in the project, and $n = (1 + \sqrt{R_1})/(1 - \sqrt{R_1})$ is substituted into the phase-spacing expression for d .

This part of the study is important because it shows that the multiple-beam result was not obtained by simply reusing the two-beam index. The interface response and the phase spacing are estimated together, subject to the same slowly varying optical-parameter assumption used earlier.

Numerical Diagnostics. The analysis provides more than the final thicknesses. For the two-beam calculation, dataset 1 has mean refractive index 2.49 with variance 1.41×10^{-4} , while dataset 2 has mean refractive index 2.55 with variance 3.61×10^{-4} . For the multiple-reflection calculation, the corresponding SiC means are 2.47 and 2.53, with variances 5.79×10^{-4} and 1.18×10^{-4} .

The reported reflectance variances are 1.15×10^{-5} , 2.29×10^{-5} , 7.85×10^{-6} , and 1.18×10^{-5} for datasets 1–4, respectively. These small within-window variations support the local-constant approximation used between neighboring extrema. They do not, however, remove systematic uncertainty in the dispersion relation or in the phase convention at the interfaces.

The two SiC estimates also provide a direct model-comparison diagnostic. Switching from the two-beam to the multiple-beam calculation changes dataset 1 from 8.50 to 8.55 μm and dataset 2 from 8.60 to 8.63 μm . The relative changes are approximately 0.59% and 0.35%, both well below the project’s 5% materiality rule. The conclusion that multiple reflections are a small correction for these two spectra therefore follows from the project’s own criterion rather than from visual similarity alone.

5 Discussion

5.1 Uncertainty and Interpretation

Three different notions of agreement appear in the source analysis and should be kept separate. First, the reported fit correlations describe how well the transformed spectral points follow the fitted relation. Second, the two incidence

angles give similar SiC thicknesses. Third, the two optical models give similar thicknesses. All three are useful internal checks, but none is an independent measurement of absolute error.

A first-order sensitivity reading of the local estimator is straightforward. Since d is inversely proportional to the extrema spacing Δs , a relative error in Δs produces approximately the opposite relative error in d when the angular and index factors are fixed. Index uncertainty enters through the optical-path factor and is systematic if the same dispersion relation is used for all extrema. Angle error is comparatively weak at the modest incidence angles used here, consistent with the small polarization difference calculated in the analysis.

The recent phase-resolved FTIR study of Li et al. treats dispersion, absorption, envelope normalization, and phase reconstruction explicitly, illustrating why a full-spectrum inversion can improve on adjacent-extrema methods when higher metrological accuracy is required.[1] Here, the emphasis is on a transparent fringe-based derivation and a cross-check between simple and multiple-reflection models using the measured spectra.

Model Behavior and Interpretation. The main strength of the calculation is that the optical assumptions remain visible. The analysis identifies a usable spectral window, checks the effect of polarization at the actual incidence angles, and compares a single-round-trip approximation with an explicit infinite reflection series. The resulting thicknesses are stable across these modeling choices.

The same transparency also shows the limitations. First, the refractive-index relation is treated as known; uncertainty in dispersion transfers directly to geometric thickness. Second, the peak-selection procedure depends on extrema identification and regularization, so the resulting thickness is sensitive to fringe selection in noisy spectral regions. Third, absorption is handled mainly by excluding the strongly distorted region rather than fitting a complex dielectric function. A more complete treatment would fit the full reflectance spectrum with a transfer-matrix or phase-resolved model. Recent SiC work follows that direction by combining background removal, phase reconstruction, Fourier initialization, and least-squares refinement in a dispersive three-layer model.[1]

Assumptions and Computational Procedure. The calculation relies on four assumptions. The interfaces are treated as sufficiently smooth and parallel for coherent interference; the beam separation after internal propagation is assumed small relative to the beam cross-section, so the interfering components overlap spatially; optical parameters are assumed to vary slowly between neighboring extrema in the selected high-wavenumber window; and absorption is treated as weak enough there for the extrema-based model to remain useful. The later multiple-reflection section relaxes the first optical simplification—the number of retained round trips—but not the assumptions of local homogeneity and coherent overlap.

The numerical procedure used for the two-beam calculation can be summarized without reproducing the source code. The spreadsheet spectra are read as wavenumber-reflectance pairs; the analysis window is restricted to 2500–4000 cm^{-1} ; extrema are identified and regularized using the approximately 126 cm^{-1} fringe interval described in the report; neighboring extrema provide y_1 , y_2 and Δs ; R_1 and n are recovered from the envelope equations; and the angular factor is evaluated at the measurement angle before computing d . The reported thickness is then obtained from multiple local estimates rather than from one isolated pair.

For the multiple-reflection calculation, the same spectra are transformed to the $s-1/(1-R)$ representation. Adjacent extrema are used to solve first for $R_1 R_2$ and then for the two interface reflectances. The air/film reflectance gives a local refractive-index estimate, after which the same fringe-spacing phase relation yields thickness. The source program also reports the mean and variance of the recovered refractive index and reflectance as diagnostics.

This procedure explains why the two models can return similar thickness while producing slightly different local optical parameters. The phase periodicity is controlled primarily by optical thickness, whereas the envelope shape and contrast determine how the interface coefficients are inferred.

Cross-Angle and Cross-Material Consistency. The first two datasets measure SiC at different incidence angles. The two-beam results, 8.50 and 8.60 μm , differ by about 1.18% relative to 8.50 μm . The analysis uses this agreement, together with the small local variance in recovered n , as evidence that the angular correction and the selected spectral window are behaving consistently.

Datasets 3 and 4 provide a second material regime. Their recovered refractive indices, 3.25 and 3.49, and reflectances, 28.0% and 30.7%, are appreciably larger than the SiC values in the source calculation. Their thicknesses, 3.79 and 3.53 μm , are correspondingly distinct from the roughly 8.6 μm SiC layer. The point of these additional datasets is not to claim that the same dispersion parameters apply to both materials, but to test the multiple-reflection algebra under a different interface-reflectance scale.

The quantitative discussion below follows the dataset-specific result table, which reports thickness, refractive index, reflectance, and fit diagnostics together and therefore provides the most internally consistent mapping between measurements and reported values.

Fresnel Check at the Larger Incidence Angle. A polarization check at the larger incidence angle provides a numerical test of the later simplification. Taking the larger measurement angle, $\theta_0 = 15^\circ$, and a representative index $n = 2.5$,

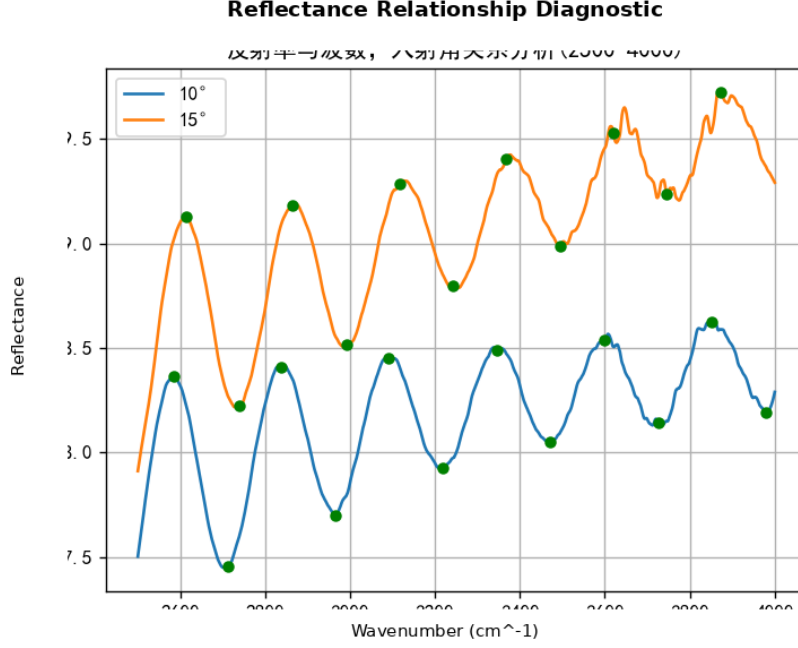


Figure 4: Diagnostic relationship plot from the SiC analysis.

the normal-incidence approximation gives

$$R = \left(\frac{n - 1}{n + 1} \right)^2 \approx 18.37\%. \quad (20)$$

Using Snell’s law to obtain the internal angle and evaluating the two Fresnel polarizations separately, the source calculation gives $R_s \approx 19.41\%$ and $R_p \approx 17.35\%$. Their unpolarized average is approximately 18.38%, essentially the same value used by the scalar approximation in the project.

The important point is not that polarization can always be ignored, but that the project checked the worst of its two measurement angles and found a sub-percent difference in the averaged reflectance used for index recovery. At larger angles, near a Brewster condition, or with a different index contrast, the same simplification would need to be reconsidered.

High-Wavenumber Window Selection. The original spectra show a pronounced disturbance near the SiC lattice-vibration region around 800–1000 cm^{-1} . In that region, the report notes rapid variation in both refractive index and extinction behavior, with reflectance excursions that can obscure the regular interference pattern. The extrema-spacing method assumes a comparatively smooth envelope and slowly varying optical parameters, so using those distorted fringes would violate the assumptions behind the estimator.

The 2500–4000 cm^{-1} interval is selected because the fringes are visibly more regular and neighboring extrema are separated by roughly 120–135 cm^{-1} . Applying the adopted dispersion relation at neighboring extrema gives a maximum local index change of about 1% in the source calculation. This is the empirical basis for treating n as locally constant over one fringe while still allowing it to change over the full fitting interval.

The analysis further notes that measured peak locations contain noise. Rather than accepting every local maximum independently, it anchors the first and last reliable extrema and uses the approximately regular spacing to stabilize the intermediate sequence. This choice trades some local spectral detail for a more stable phase-order assignment. A more advanced phase-unwrapping method could avoid that regularization and would provide a useful comparison with the extrema-based procedure.

6 Conclusion

The infrared-fringe model yields SiC epitaxial thicknesses near 8.5–8.6 μm for both measured incidence angles. The repeated-reflection calculation changes the result only slightly for the measured spectra, while the two silicon-wafer datasets give thicknesses near 3.5–3.8 μm . The evidence supports consistency of the optical model across angles and interference treatments. It does not, by itself, establish traceable absolute accuracy; that would require independently calibrated thicknesses and a full uncertainty analysis.

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